

Computer Security: Practical Network Attack Simulation Tool

❖ **Objectives of the Practical Assignment :**

Develop a **network attack simulation tool** with a graphical user interface (GUI). The tool should allow users to choose from several attack types, configure their parameters, and launch the attacks. All work must be done ethically and tested only in a controlled environment.

❖ **Technical Requirements:**

1. **Technologies to Use:**
 - **Backend:** Python with *Scapy* for packet manipulation.
 - **Graphical Interface:** *Tkinter* or *PyQt* or ...
2. **Attacks to Implement:**
 - **SYN Flood.**
 - **UDP Flood.**
 - **ICMP Flood.**
 - **HTTP Flood.**
 - **Synonymous Flood.**
 - **ARP Poisoning:** Create an attack that poisons the ARP table by sending *who-has* packets.
3. **Graphical Interface Features:**
 - The user should **first select the type of attack** from a menu or list.
 - After selecting the attack, input fields for configuration should appear (e.g., target IP, target port, number of packets, etc.).
 - Include buttons to **start/stop each attack** and a log display area to show progress or errors.
4. **Additional Requirements:**
 - **Source addresses must be random** if the attack requires it (e.g., SYN Flood, UDP Flood).
 - Work must be done in **pairs**.
 - You are free to **innovate** and add extra features or improvements if you want.
 - The **project deadline is January 8, 2025** (the week after the holidays). **Validation will be done then, inshallah.**

❖ **Evaluation Criteria:**

- Backend Implementation: 60%.
- Functionality and Design of the GUI: 30%.
- Creativity and Optional Features: 10%.